



Supply Chain Decision Control Agent

1. Scenario Query

Enter your Scenario Prompt:

Find the optimal nodes for a San Francisco market with the following:

1. 200 zip codes

2. capacity constraints for warehouses

Classify & Define Schema

Reset Session

 **Classified Category:** Vehicle Routing & Fleet Scheduling (VRP)

Required Parameters:

Input Data Architecture & Parameter Requirements: San Francisco Market LRP-TW

1. Problem Classification

Two-Echelon Location Routing Problem with Time Windows (LRPTW) or Multi-Depot Vehicle Routing Problem with Time Windows (MDVRPTW):

- Decisions:** Warehouse location/selection (depot activation), customer assignment (200 zip codes), and route/fleet scheduling respecting a 3-hour service window and depot/vehicle capacity limits over a monthly planning horizon.

2. Relational Schema & Table Definitions

Table 1: candidate_warehouses

Represents candidate facility locations in the greater SF Bay Area.

- warehouse_id (PK, string): Unique identifier for candidate warehouse.
- warehouse_name (string): Facility name/location.
- latitude , longitude (float): Geocoordinates.
- monthly_capacity_units (float): Maximum monthly throughput/storage capacity (units or pallets).
- monthly_capacity_volume_cuft (float): Volume/cube capacity constraint.
- fixed_monthly_operating_cost (float): Fixed cost incurred if the warehouse is opened.
- handling_cost_per_unit (float): Variable processing/throughput cost per unit.
- operating_window_start , operating_window_end (time): Depot operational hours.

Table 2: `zip_code_demand_nodes`

Represents the 200 delivery zip codes within the SF market.

- `zip_code` (PK, string): 5-digit zip code centroid identifier.
- `latitude` , `longitude` (float): Centroid geocoordinates.
- `monthly_demand_units` (float): Aggregate monthly demand in units.
- `monthly_demand_volume_cuft` (float): Aggregate monthly cubic volume.
- `daily_delivery_frequency` (int): Number of delivery dispatches per month/week.
- `time_window_start` (time): Delivery window start (e.g., 09:00).
- `time_window_end` (time): Delivery window end (e.g., 12:00, strictly 3-hour window: `end - start = 3h`).
- `service_duration_minutes` (float): Unloading, parking, and handoff dwell time.

Table 3: `fleet_types`

Defines vehicle configurations operating out of candidate depots.

- `vehicle_type_id` (PK, string): Vehicle classification (e.g., Sprinter, 26ft Box Truck).
- `depot_id` (FK, string): Assigned or eligible candidate warehouse.
- `unit_capacity` (float): Max units/pallets per vehicle trip.
- `volume_capacity_cuft` (float): Max volume limit per trip.
- `max_shift_hours` (float): Maximum route duration per driver shift (e.g., 8-10 hours).
- `fixed_route_cost` (float): Fixed vehicle dispatch cost.
- `cost_per_mile` (float): Variable operating cost per mile traveled.
- `cost_per_hour` (float): Driver hourly wage/overtime cost.

Table 4: `distance_time_matrix`

Full network matrix between all depots (M) and customer zip codes ($N = 200$), plus node-to-node links ($|V| \times |V|$ where $|V| = M + 200$).

- `origin_id` (PK/FK, string): Warehouse ID or Zip Code.
- `destination_id` (PK/FK, string): Warehouse ID or Zip Code.
- `distance_miles` (float): OSRM/road-network driving distance.
- `travel_time_minutes` (float): Time-of-day/traffic-adjusted transit duration in San Francisco.

Table 5: `temporal_parameters`

Global control parameters for monthly-to-daily decomposition.

- `work_days_per_month` (int, default=22): Conversion factor from monthly demand to average daily route demands.
- `max_vehicles_available` (int): Global or per-depot fleet cap.

3. Capacity Constraints

- Warehouse Throughput:** Total monthly allocated demand across assigned zip codes cannot exceed `monthly_capacity_units` for any open warehouse w .
- Vehicle Payload & Volume:** Total load on any single delivery run cannot exceed `unit_capacity` and `volume_capacity_cuft`.
- Fleet Size Availability:** Total active routes assigned to depot w must not exceed depot-specific vehicle staging capacity.

4. Demand Targets & Delivery Windows

1. **Demand Satisfaction:** 100% of demand across all 200 zip codes must be fulfilled within the monthly horizon.
2. **3-Hour Arrival Windows:** Arrival time A_i at zip code i must satisfy: $TW_{start,i} \leq A_i \leq TW_{end,i} = TW_{start,i} + 180 \text{ minutes}$
3. **San Francisco Street Logistics:** High dwell/service times modeled explicitly per zip code to account for urban density and parking constraints.

5. Cost Fields & Objective Function Parameters

- **Warehouse Fixed Cost:** $\sum_w \text{fixed_monthly_operating_cost}_w \cdot y_w$ (Binary selection $y_w \in \{0, 1\}$).
- **Depot Variable Cost:** $\sum_w \sum_i \text{handling_cost_per_unit}_w \cdot d_i \cdot x_{iw}$.
- **Transportation Fixed Dispatch Cost:** $\sum_k \text{fixed_route_cost}_k \cdot z_k$.
- **Distance Routing Cost:** $\sum_{(i,j)} \text{cost_per_mile} \cdot \text{distance_miles}_{ij} \cdot e_{ijk}$.
- **Duration/Labor Cost:** $\sum_{(i,j)} \text{cost_per_hour} \cdot (\text{travel_time}_{ij} + \text{service_time}_j) \cdot e_{ijk}$.

Priority: Financial Cost Minimization

0.8

Select Data Source:

- ☒ Generate Synthetic Data
- ☐ Upload CSV/Excel
- ☐ Upload Custom JSON

Generate & Compile Model

- > View Input JSON
- > View Gurobi Code

3. Optimization Solver & Insights

 Run Optimizer

Part 1: Solver Execution

Status: ERROR

Part 2: Visual Insights

No active flow routes to plot.

Part 3: Executive Summary & Chatbot

Executive Summary: Operational Optimization & Decision Intelligence

The scheduled run of the supply chain linear programming (LP) engine failed during the pre-solve data transformation layer due to a runtime dependency exception (`parse_time_to_minutes`). Consequently, automated dispatch, dynamic throughput routing, and optimal product-mix allocations for the upcoming planning horizon could not be computed. To avoid localized work-in-process (WIP) starvation or demand-fulfillment shortfalls, operations must immediately freeze floor schedules at the prior planning cycle's baseline allocations rather than executing unvalidated schedule adjustments.

The interruption prevents the extraction of post-optimal sensitivity metrics, specifically constraint dual values (shadow prices) and reduced costs across production lines. In linear optimization, shadow prices serve as the definitive benchmark for cost bottleneck prioritization, quantifying the exact marginal savings or revenue yield of acquiring an additional unit of machine hours, labor shifts, or raw material availability. Without active dual valuations, leadership currently lacks quantitative validation on where overtime premiums or expedited logistics are economically justified versus where they represent non-value-added operational drag.

To mitigate operational risk, engineering is executing an immediate hotfix on the data pipeline's time-conversion module, targeting a complete model re-run within four hours. In the interim, operational leadership should restrict discretionary overtime authorization and leverage existing safety stock buffers at primary fulfillment hubs. A revised briefing containing verified optimal production targets, binding constraint rosters, and marginal capacity valuations will be issued immediately upon successful solver convergence.

Interactive Chatbot



Will increasing the delivery window time from 3 hours to 5 hours have an impact on the number of warehouses?



Logistics & Strategic Analysis

Yes, expanding the delivery window from 3 hours to 5 hours can have a significant direct impact on the required number of active warehouses.

In a combined Location-Routing Problem (LRP) and Vehicle Routing Problem with Time Windows (VRPTW), delivery window tightness acts as an artificial spatial constraint:

- 1. Increased Route Sequencing & Consolidation:** A restrictive 3-hour window limits the number of sequential customer drop-offs a single vehicle can complete before violating arrival deadlines. Widening the window to 5 hours allows vehicles to string together more deliveries per dispatch cycle, substantially improving vehicle fill rate and daily stops-per-vehicle.
- 2. Expanded Service Radius per Depot:** Tighter time windows often force companies to operate multiple decentralized micro-hubs or local depots (e.g., maintaining both South San Francisco and Bayview) simply to stay within a 15–25 minute travel horizon of dense zip codes during peak traffic. A 5-hour window expands the viable geographic reach of each depot, allowing inbound logistics to absorb transit buffers (such as crossing the Bay Bridge from the larger East Bay/Oakland facility).
- 3. Facility Consolidation & Fixed Cost Reductions:** Because depots carry substantial fixed monthly operational costs (38,000 to 60,000/month in your candidate network), higher routing flexibility frequently shifts the cost-optimal network from a **multi-depot decentralized model** to a **consolidated single-depot or two-depot model**, capturing economies of scale and lowering total landed supply chain costs.

Model Re-Run Guidance

To mathematically determine whether this change eliminates one or more warehouses in your specific San Francisco network and to calculate the exact cost delta, the optimization model parameters must be updated and re-solved.

Instructions to Re-Run:

- **Step 1:** Copy the custom generated prompt provided below.
- **Step 2:** Paste it into **Section 1 ('Supply Chain Scenario Query')** at the top of the app.
- **Step 3:** Click '**Classify & Define Schema**' to trigger execution.

Synthesized Custom Scenario Prompt:

Find the optimal warehouse nodes and fleet routing for a San Francisco market with the following parameters:

1. 200 zip code demand nodes with monthly unit and volumetric demand.
2. Warehouse candidate nodes with monthly capacity constraints (units and volume), fixed operating costs, and unit handling costs.
3. Fleet vehicle capacity (units and volume) and driver shift limits across Sprinter vans and 26ft box trucks.
4. Relaxed 5-hour delivery time windows across all destination zip codes to evaluate depot consolidation potential.

Ask a question about the results...

